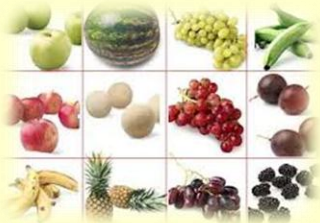


Introduction to Classification



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Classification: Definition

- Given a collection of records (*training set*)
 - Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
 - A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

Tan, Steinbach, Kumar, *Introduction to Data Mining*

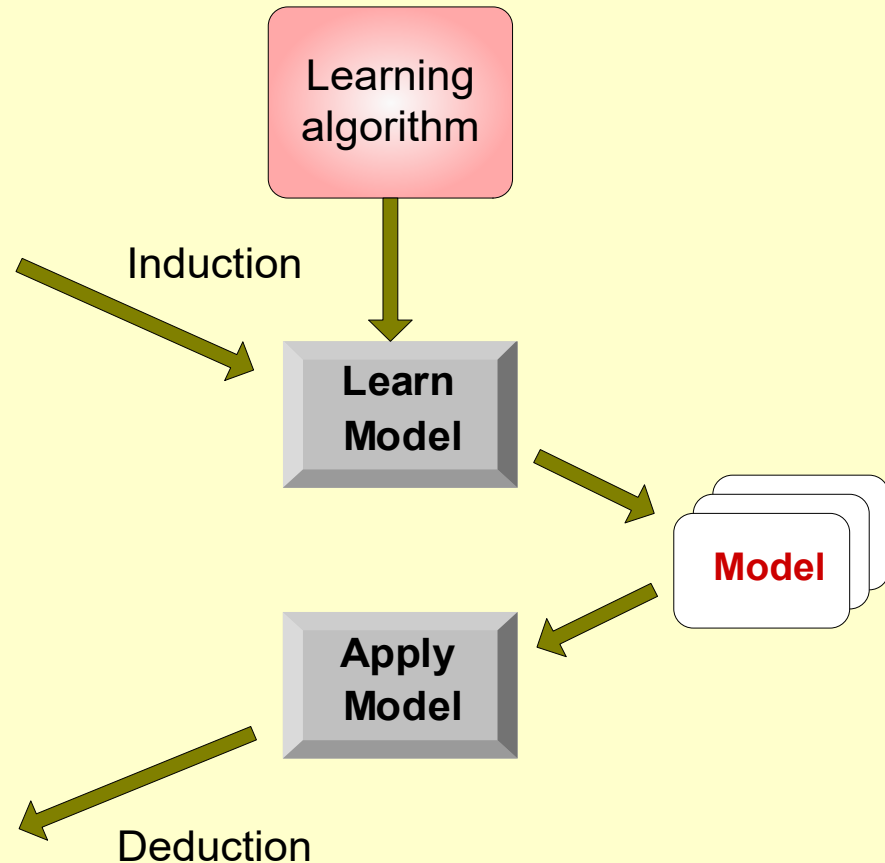
Illustrating Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

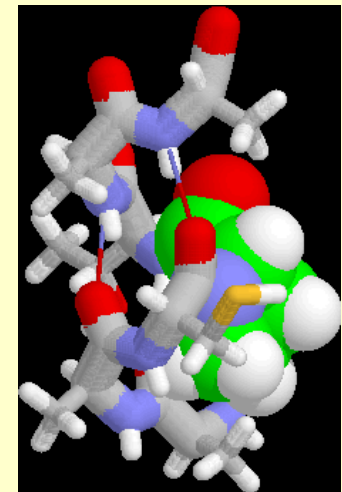
Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Examples of Classification Task

- Predicting tumor cells as benign or malignant
- Classifying credit card transactions as legitimate or fraudulent
- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil
- Categorizing news stories as finance, weather, entertainment, sports, etc



Steps of Classification

- The classification process is divided into three steps: learning and testing.
 - Learning → Some data those have class label is induced to build a model of generalization
 - Validation → the model is examined with some of other data those have hidden class label, in order to determine accuracy of the model
 - Testing → the model is examined with new data those have no class label
- However, in the context classification performance, the validation step is commonly mentioned as testing
- If the accuracy is sufficient, this model can be used to predict unknown data classes.
- Classification is characterized by training data that has a label, based on this label the classification process obtains a pattern of attributes from a data.

Classification using Nearest Neighbors (NN)

- A simple method to classify a new data based on similarity with labeled data
- Similarity usually uses the distance metric
- The unit of distance generally uses the Euclidian
- Has several names: lazy algorithm, memory-based, instance-based, exemplar-based, case-based, experience-based

Types of NN

- 1-NN
 - Classification is based on 1 nearest training data
- k -NN
 - Classification is based on several (k) nearest training data. The classification result is determined with voting from class labels of k nearest training data
 - $k > 1$

1-NN Algorithm

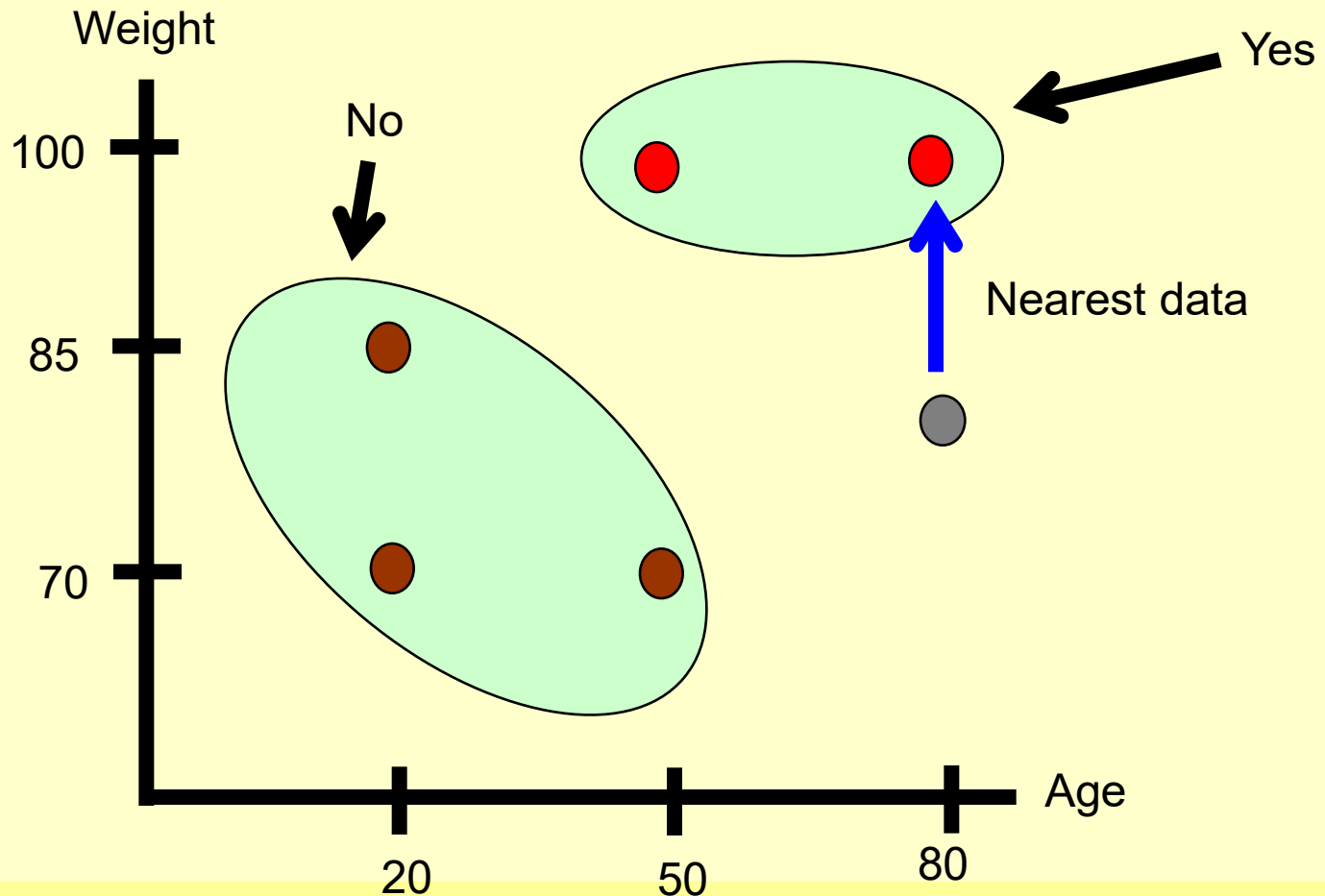
- Calculate the distance between a new data to training data
- Determine 1 nearest training data
- Classify a new data into the label of the 1 nearest training data

Example

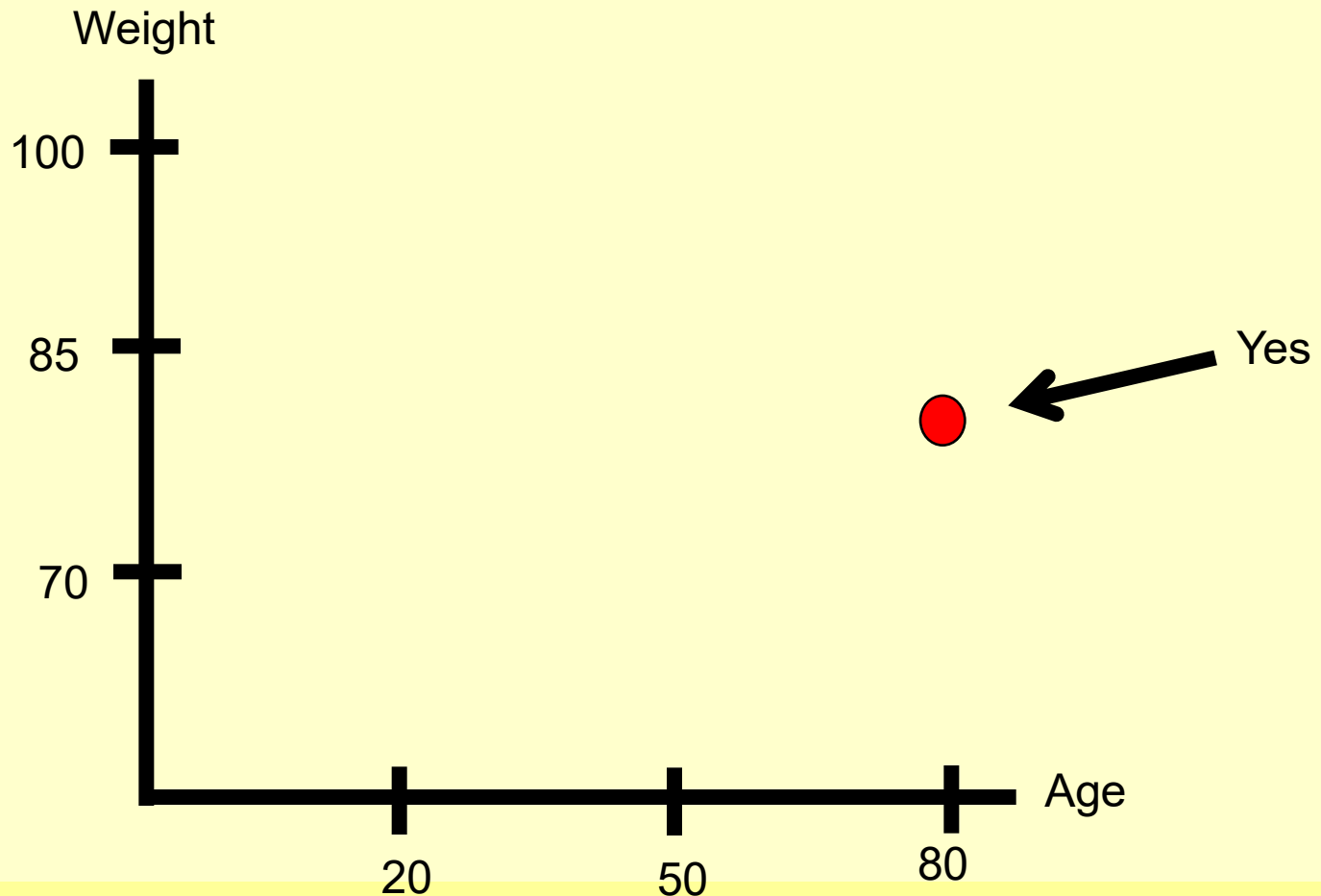
Age	Weight	Hypertension
20	70	No
20	85	No
50	70	No
50	100	Yes
80	100	Yes
80	85	?

← a new data

Classification with 1-NN



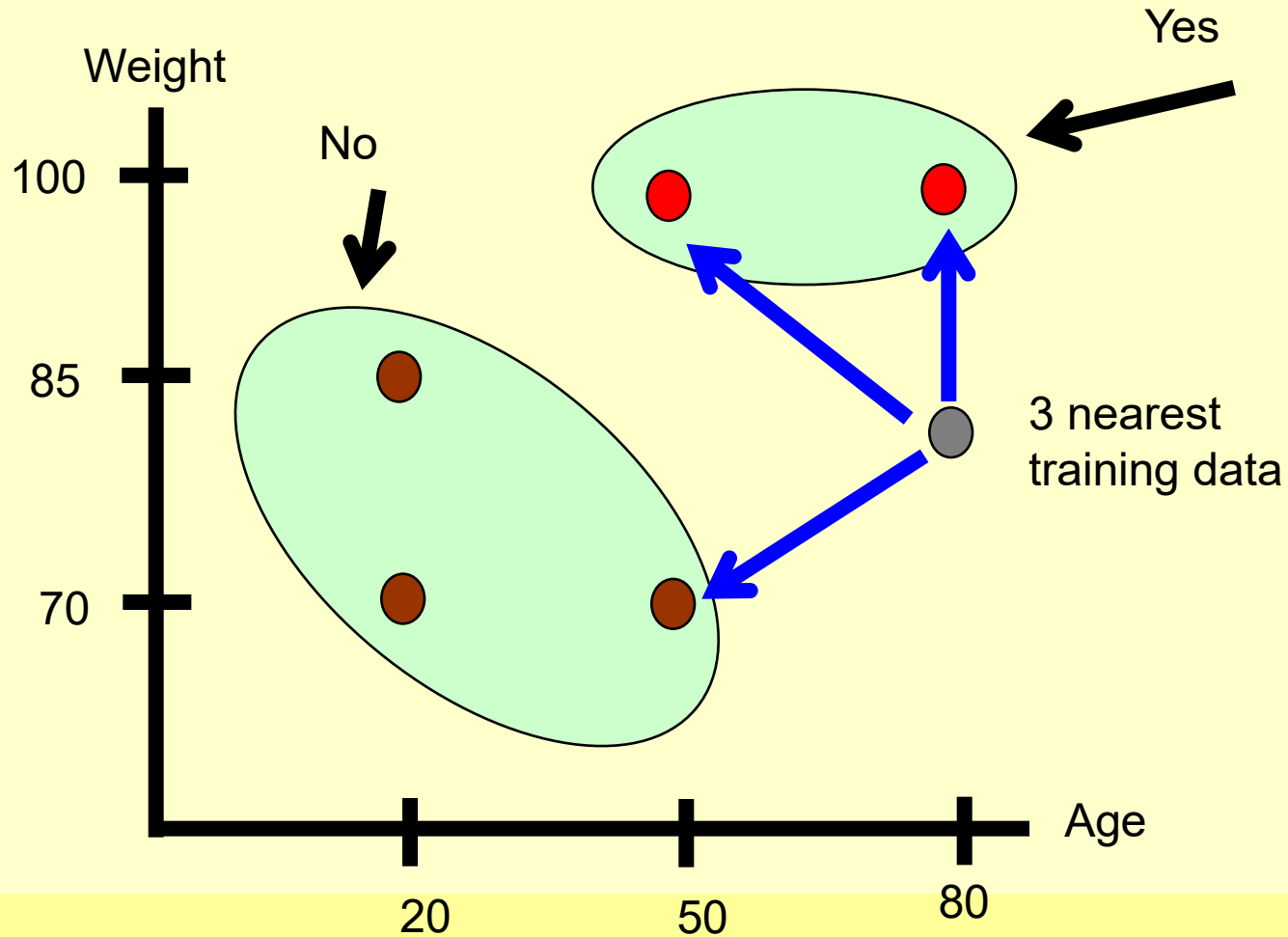
Classification with 1-NN



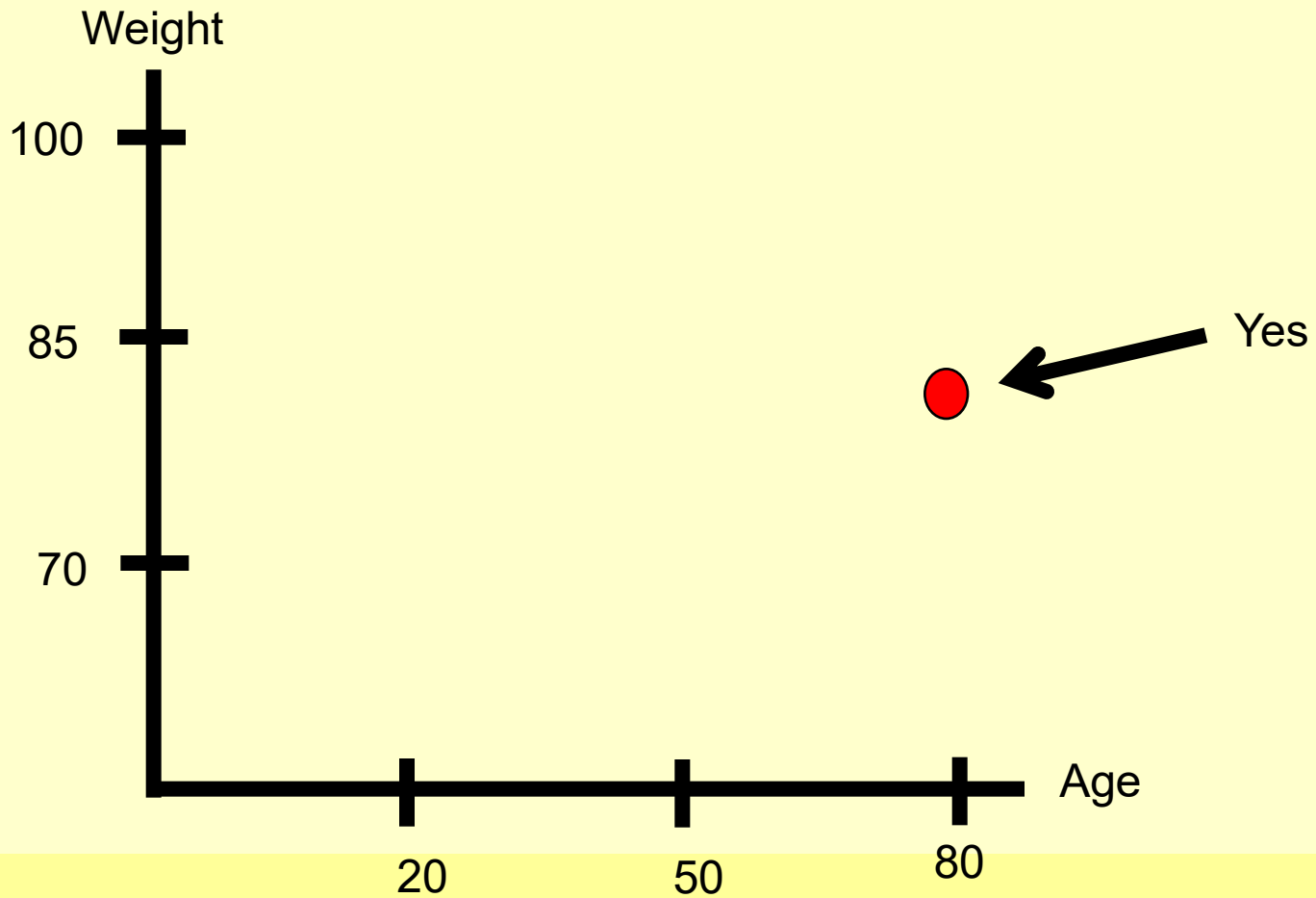
k-NN Algorithm

- Determine k
- Calculate the distance between a new data to all training data
- Find k nearest training data
- Vote class labels of the k nearest training data and classify a new data into the winning vote class label

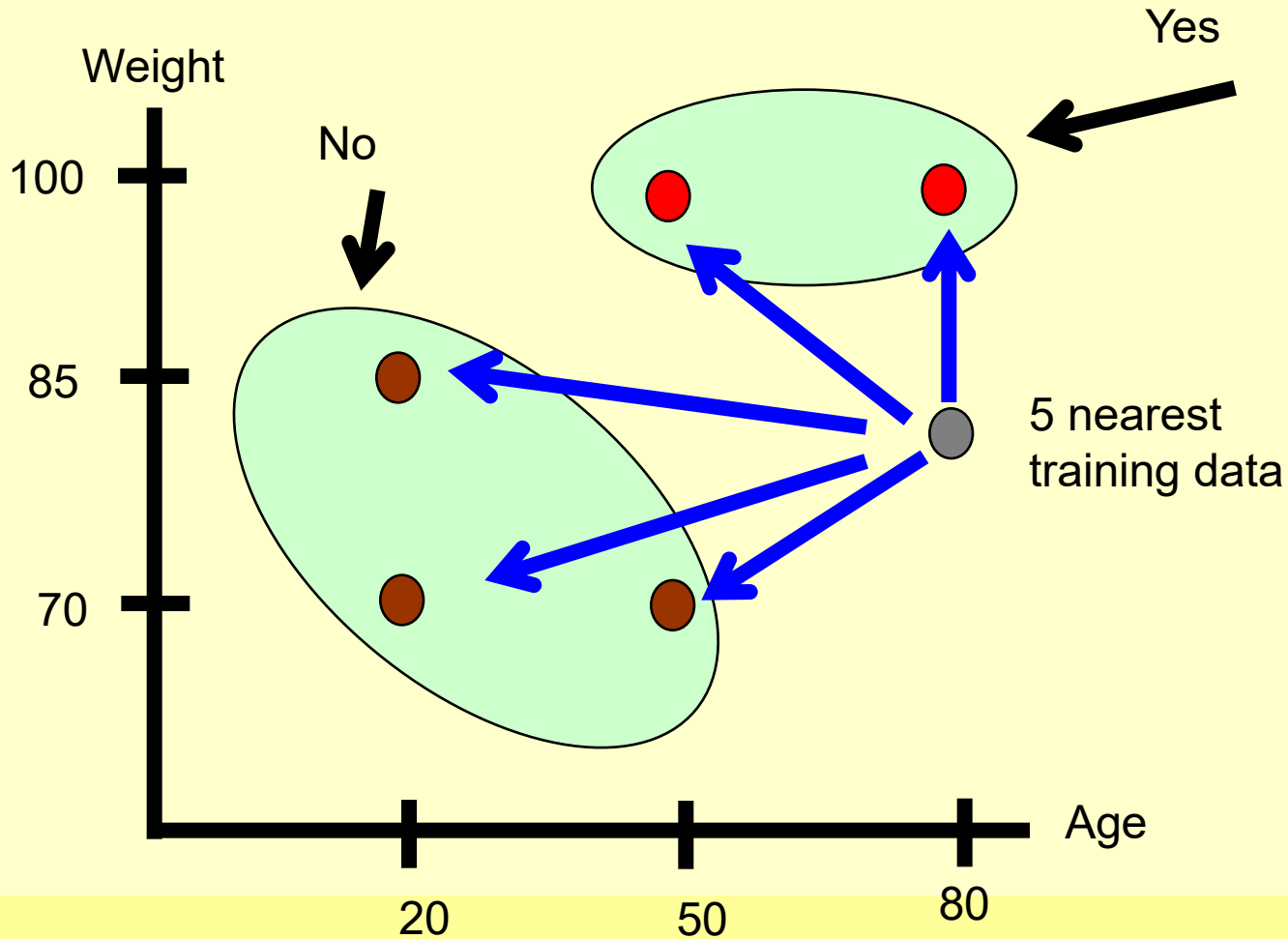
For example, $k=3$



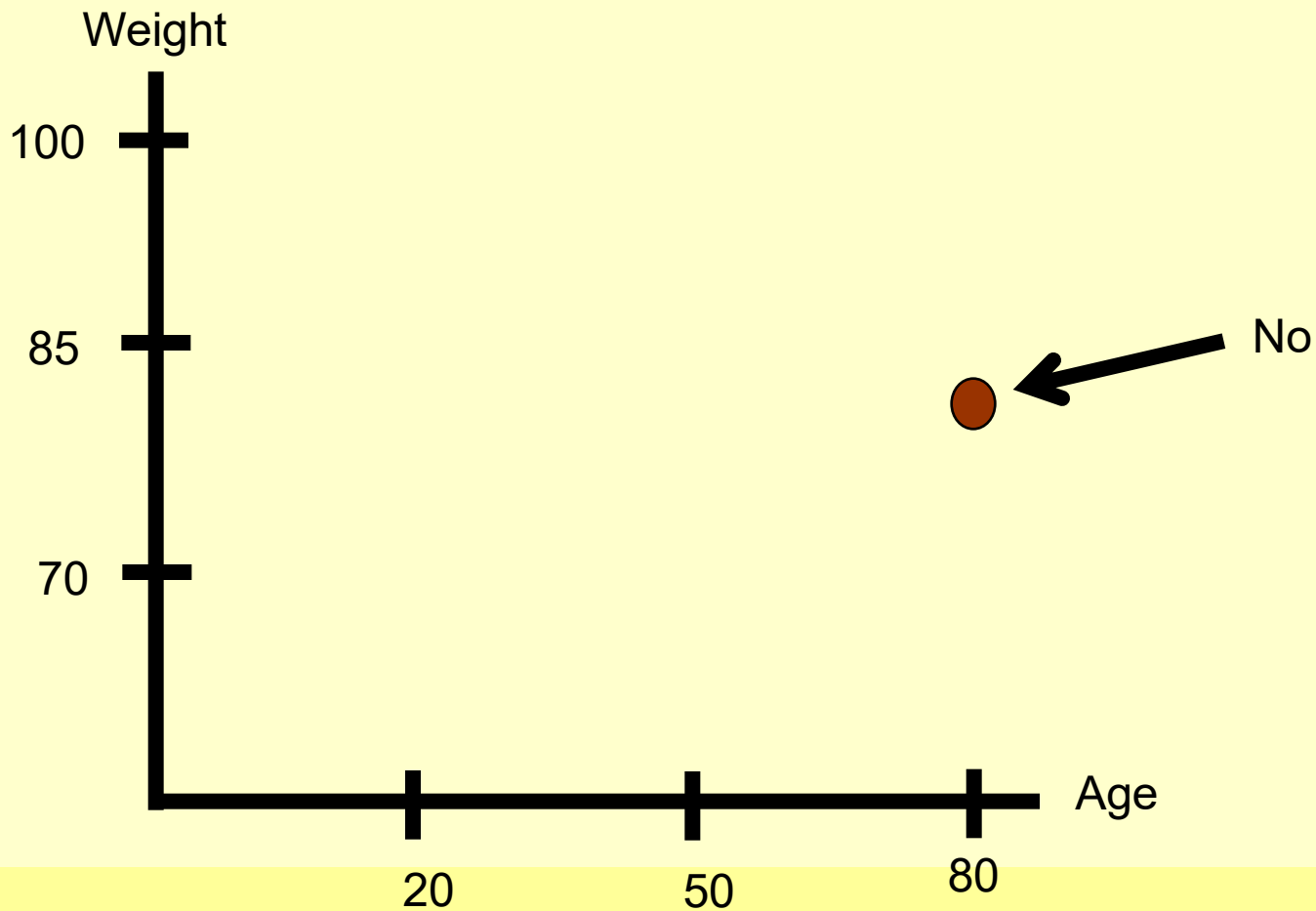
For example, $k=3$



For example, $k=5$



For example, $k=5$



- Advantage:
 - Simple and analytically tractable
 - Possible to apply parallel implementation
 - Error rate: $>$ bayesian, $<$ 2xbayesian
- Disadvantage:
 - Need huge memory and computation